# Administrivia

1

Pre

CS 3244
Machine Learning

NUS | Computing
National University of Singapore

# Above the Hood

- "Big Picture" applications
- Set up and go
- Hands-on exercises
- Connection with Industry
- Mock Job Interview

For executive and continuing lifelong education.

# Below the Hood

- Mathematical Details
- Proofs

The math is important for mastery and grounding insight, but most of you will forget the details in the long run.

# Barriers to success in this course

- Serious Math.
  - Probability and Statistics
  - Linear Algebra
- Programming competence.
- Flipped course implementation – "The New Normal"

Will need your support and understanding – please offer feedback to us too.

# Student Learning Outcomes

1. Understand the basic concepts of machine learning

2. Apply an appropriate machine learning algorithm for a given problem.

3. Evaluate the performance of a machine learning solution.

4. Use a machine learning tool to carry out machine learning experiments

# Python & Jupyter

We'll be using the Python programming language for our entire class homework assignments, in particular
the 3.6.x version.

We'll also be using Jupyter / iPython Notebooks to show and simulate experiments.

Google has a very nice, convenient Python-in-a-browser experience, Colab, which we'll take advantage of.

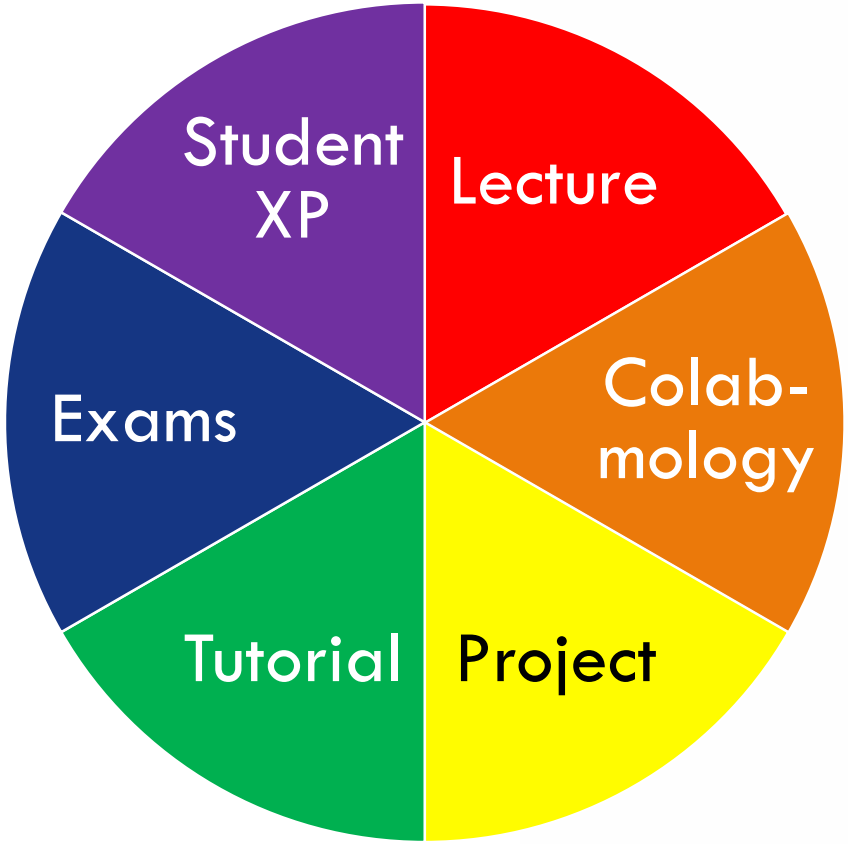
Python is an easy language to transition to for most programmers and it has numerous libraries that work well with machine learning as well as data analysis.

# Course Components

## CS3244 Machine Learning

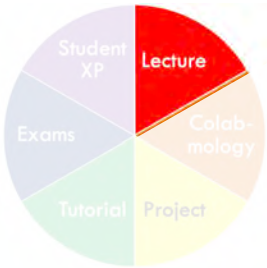
NUS National University of Singapore
Department of Computer Science
School of Computing

# Components

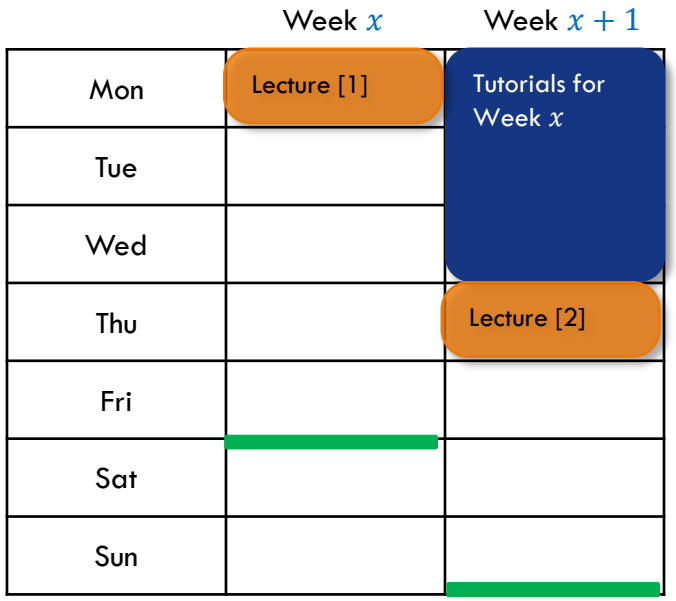



More details on the CS3244 website

# 1. Lectures – Flipped Classroom




*(Study when and where you want)*




*Every week you need to:*

1. Watch the Week $x$ Pre videos and complete the Pre Assessment by Fri 23:59;
2. Attend the Week $x$ corresponding tutorial Mon-Wed of the following week;
3. Optionally attend Week $x$ corresponding help session, after tutorials;
4. Complete the Week $x$ Post videos and complete the Post Assessment by Sun 23:59

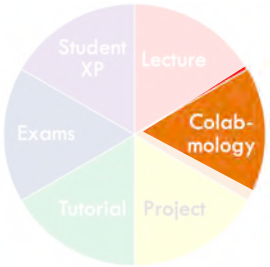
# Optional help sessions via Zoom

For those wanting a bit more guidance, come to our help sessions during the normal class times online in Zoom (Weeks 02–06 and 08–12; Mondays 16:00-18:00).

Purely voluntary. We'll be recapping what was covered in the previous lecture, tutorial first. Then cover your questions and get to know you better.

# 2. Colabmology

*(Practice as much as you need)*




Google Colab Notebooks

- Interactive programming practice
- On your own; no stakes

Coursemology Learning Management System

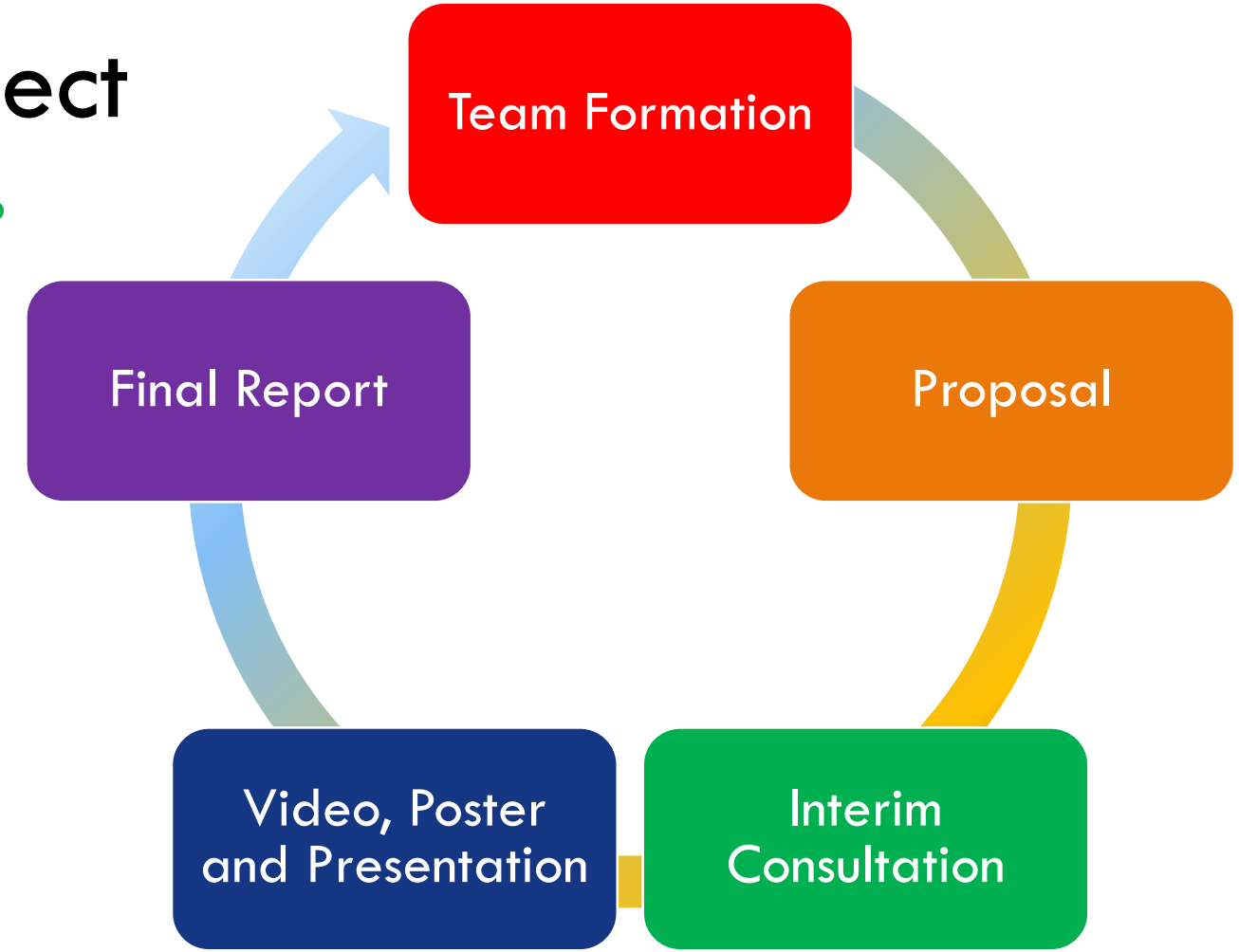
- All official information here (not on the external website)
- Gamified
- Auto grading of exercises that you do in Colab
- Your TAs will be monitoring your progress

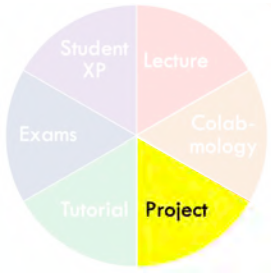
LumiNUS will still hold your official grades

10% of your grade in total

# 3. Project

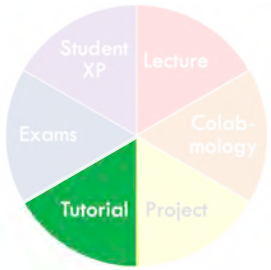
*(In-depth, group self-determined immersive project)*




35% of your grade in total

# 4. Tutorials via Zoom



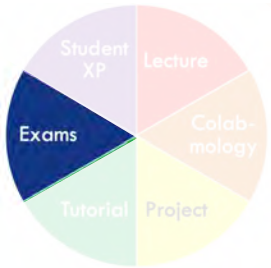

*(Face to face check on your understanding in groups)*

Tutorials start in Week 03 and will be entirely online in Zoom.

Have another device (e.g., phone) ready for warm-up exercises.

| | 1000 | 1100 | 1200 | 1300 | 1400 | 1500 | 1600 | 1700 | 1800 |
|---|---|---|---|---|---|---|---|---|---|
| Mon | | | Tutorial [06] | Tutorial [01] | Tutorial [04] | Tutorial [03] | Lecture [1] | | |
| Tue | | | | | | Tutorial [09] | Tutorial [10] | | |
| Wed | Tutorial [08] | Tutorial [05] | | | | Tutorial [07] | Tutorial [02] | | |
| Thu | | Lecture [2] | | | | | | | |
| Fri | | | | | | | | | |

# 5. Exams




*(Primary assessment of the basic concepts)*

E-Exam 1 (25% of final grade; Week 07, likely just Mon)

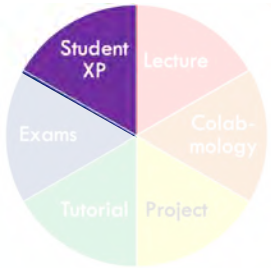
- Weeks 01–06

E-Exam 2 (25%) of final grade; Week 13, both Mon & Thu)

- Emphasis on Weeks 07–12, but may cover earlier parts of the course as well

Both open book

# 6. Student XP




*(Keeping tabs on your personalized progress in the course)*

Weekly customized, emailed announcements from your lecturer, tutorial and project teaching assistants.

Clear student learning outcomes for each component and subcomponent.

Interviews with Machine Learning experts in our school and community.

# Course Policy

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# Freedom of Information Rule

Collaboration is acceptable and encouraged.

To assure that all collaboration is on the level,
**you must always fill in the name(s) of your collaborators on your assignments and project work.**

You will be assessed for the parts for which you claim is your own contribution.

# Pokémon Go

You are free to meet with fellow students(s) and discuss assignments with them.

Writing on a board or shared piece of paper is acceptable during the meeting; however, you **may not take any written (electronic or otherwise) record away from the meeting**.

After the meeting, do something else for at least a half-hour (Pokémon Go, or doing an assignment for a different class), before working on the assignment.

> This will assure that you are able to reconstruct what you learned from the meeting, by yourself, using your own brain.
>
> You will be asked to certify that you meet this requirement per assessment.

# Online Procedures

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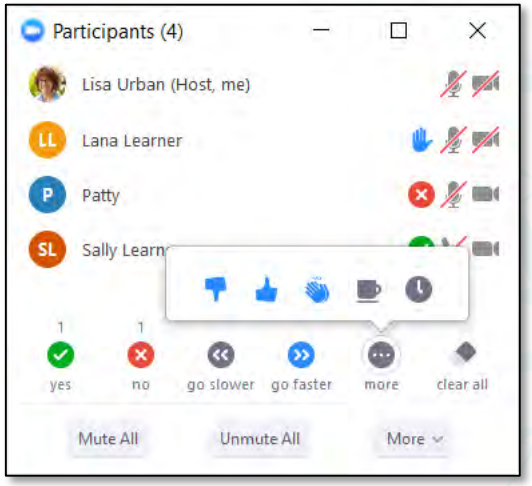
# Zoom Etiquette for Tutorial




Each tutorial will have a tutorial leader and a backup.

Have your video on where possible.

Leaders will be giving the tutorial, while the backup monitors participation and chat.

Be familiar with polling, changing your name, screen annotation, and chat.

There will be a sample, familiarization run during Week 02.

# Zoom Exam Procedure

2× 1 ½ hour exams in Weeks 07 and 13.
Each exam is multipart, where you may not revisit the previous parts.
Short washroom break between parts.

Proctoring by Zoom and by ffmpeg desktop recording.

- Zoom: set up a side camera or external webcam and connect to the proctor.
- ffmpeg: record your single screen desktop and upload to LumiNUS.

There will be a mandatory familiarization run during Week 05.